Agricultural Supply Chain Tracking: Track the movement of agricultural products from farm to market using GPS and RFID technology.

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***Abstract*—In the modern agri-food industry, transparency and traceability across the agricultural supply chain are paramount for ensuring food safety, reducing waste, and improving logistics. Traditional supply chains lack real-time visibility and are prone to inefficiencies, leading to increased spoilage, cost, and diminished consumer trust. This paper presents a GPS and RFID-enabled smart tracking system that monitors the movement of agricultural products from farm to market. By integrating Internet of Things (IoT) technologies such as RFID tags, GPS modules, and cloud-based data management systems, the solution ensures accurate, real-time tracking of product location and storage conditions. The system offers automated data logging, environmental condition monitoring during transit, and digital traceability accessible via a user-friendly interface. Results show that the proposed solution significantly enhances supply chain efficiency, reduces food spoilage, and supports decision-making through real-time analytics. Future work explores integration with blockchain for immutable data integrity and machine learning models for predictive insights.**

***Keywords*—Agricultural Supply Chain, RFID, GPS Tracking, IoT, Real-Time Monitoring, Food Traceability, Smart Logistics**

I. Introduction

The global agricultural supply chain plays a critical role in connecting farms with markets, ensuring that fresh produce reaches consumers in a timely and safe manner. However, this supply chain is often plagued by inefficiencies, limited visibility, and inadequate monitoring of environmental conditions, especially during transit. These challenges lead to spoilage, food waste, and a lack of accountability problems that are particularly detrimental in the case of perishable commodities such as fruits and vegetables.To address these issues, emerging technologies like the Internet of Things (IoT), Radio Frequency Identification (RFID), and Global Positioning System (GPS) tracking are being leveraged to enhance traceability and operational efficiency. RFID tags offer a reliable mechanism for identifying and tracking individual product batches, while GPS modules enable continuous geolocation tracking. When integrated into a unified IoT framework, these technologies can offer end-to-end visibility, helping stakeholders monitor product conditions, location, and status in real-time.This project proposes a smart agricultural supply chain tracking system that combines RFID and GPS technologies with cloud-based analytics to monitor the movement of goods from farm to market. The system ensures traceability by recording critical parameters such as temperature, humidity, and product location, and making the data accessible through web and mobile interfaces. Our research explores both the technological architecture and the practical implementation of this system, aiming to demonstrate how real-time data monitoring can transform traditional agricultural logistics into a more transparent, efficient, and sustainable process.

II. Literature Review

The rapid evolution of Internet of Things (IoT), wireless sensing, and automation technologies has spurred the development of numerous intelligent systems aimed at enhancing supply chain visibility and food traceability. Several research efforts and technological implementations have attempted to solve specific challenges related to real-time monitoring, product authentication, and environmental control throughout the agricultural logistics lifecycle. In [1], a smart traceability model was developed using RFID and WSN (Wireless Sensor Network) technologies to track agricultural produce through multiple supply chain stages. The system focused on monitoring temperature and location but lacked integration with cloud analytics and mobile interfaces, limiting its real-time decision-making capabilities. Similarly, a cold-chain monitoring solution presented in [2] utilized low-power sensors

to collect environmental data during storage and transport. However, the absence of GPS tracking constrained its ability to monitor product movement dynamically across geographic locations. Research in [3] introduced a chipless RFID sensor system to monitor humidity levels in packaged foods. While this approach was cost-effective and highly specific to moisture detection, it did not provide comprehensive environmental monitoring (e.g., temperature and pressure) or support location-based tracking. Other works, such as [4], explored blockchain-integrated IoT platforms to enhance food safety and traceability by recording tamper-proof supply chain events. Despite offering high data integrity, these systems posed significant overhead in terms of computational cost and infrastructure requirements. A number of studies have also explored GSM and LPWAN (Low-Power Wide-Area Network) communication modules for rural supply chain environments [5][6]. These networks offered improved coverage compared to Wi-Fi-based systems but often suffered from limited bandwidth and increased latency, making them less suitable for frequent data transmission in high-resolution tracking systems. Most existing systems either prioritize data acquisition or transmission but seldom deliver a holistic integration of GPS tracking, RFID-based inventory control, and environmental monitoring with cloud-based analytics and real-time visualization. As summarized in our system comparison, the current solutions are either costly, fragmented, or impractical for scalable deployment in developing agricultural contexts. Therefore, this project proposes a balanced, modular, and cost-efficient IoT-based tracking system that combines GPS, RFID, and environmental sensors with cloud synchronization and mobile/web dashboard access. By addressing the limitations observed in prior work, the system offers an end-to-end solution for enhancing traceability, ensuring produce quality, and minimizing losses in agricultural logistics.

III. Problem Statement

One of the pressing challenges in agricultural logistics is the lack of end-to-end visibility and real-time monitoring in the supply chain, particularly for perishable products. Traditional supply chains rely heavily on manual documentation and delayed reporting, which fail to capture the dynamic nature of product movement and condition during transportation. This results in increased instances of spoilage, food loss, and consumer distrust due to uncertainty about product origin and handling practices. While solutions like cold storage and manual inspection exist, they often fall short due to limited scalability, high operational costs, and the inability to provide real-time insights. The absence of automated tracking and environmental monitoring makes it difficult to ensure consistent quality and safety from the farm to the consumer. Although technologies such as RFID and GPS are widely available, their integration into a cohesive, IoT-enabled supply chain management system remains limited, especially in resource-constrained agricultural contexts. Existing systems either lack flexibility or are not economically viable for small to medium-scale farmers and distributors. Therefore, there is a critical need for a robust, scalable, and cost-effective smart tracking system that can provide continuous traceability of agricultural products throughout the supply chain. Such a system should utilize RFID for identification, GPS for geolocation, and IoT sensors for environmental monitoring, all integrated into a cloud-based platform for real-time access and decision-making. This project aims to fill this gap by developing and testing a prototype that enhances transparency, reduces waste, and ensures the safe and timely delivery of fresh produce.

IV. Methodology

A. An overview

The proposed system focuses on developing a smart agricultural supply chain tracking solution that monitors the environmental conditions of agricultural products and tracks their movement from the farm to the market. The system integrates temperature, humidity, and air pressure monitoring along with RFID-based inventory management and GPS-based live location tracking. The overall functionality is illustrated in the block diagram shown in Fig. 1.

The primary goal is to ensure real-time traceability of perishable agricultural goods to maintain quality, reduce spoilage, and enhance transparency across the supply chain. The monitoring setup consists of two ESP32 microcontrollers managing different tasks, including sensor data collection, RFID reading, GPS tracking, and cloud communication.

The BME280 sensor is employed to measure key environmental parameters such as temperature, humidity, and air pressure inside storage and transport environments. The RFID RC522 module manages product identification and inventory logging at each critical node (farm, transport, market). The GPS Neo-6M module provides live location updates, ensuring stakeholders are constantly aware of the cargo's position.

The ESP32 microcontrollers serve as the core processing units, interfacing with all sensors and modules. Data from the sensors and RFID modules are collected and stored temporarily in a MicroSD card module for backup purposes. Simultaneously, data is transmitted wirelessly via the ESP32 Wi-Fi module to a cloud-based monitoring platform. This enables remote access to live environmental conditions, inventory status, and location tracking through mobile and web applications.

The system's modular architecture ensures that any deviation from optimal storage or transit conditions can be detected immediately, allowing timely interventions to maintain product quality throughout the supply chain journey.

B. Schematic Design

The detailed wiring and interconnection of components are shown in the circuit diagram in Fig. 2.

The RFID RC522 module is interfaced with the first ESP32 microcontroller using the SPI communication protocol, enabling fast and reliable reading of RFID-tagged agricultural products. This ensures seamless product identification and inventory logging at the source and destination points.

The second ESP32 microcontroller handles environmental monitoring and location tracking. It connects to the BME280

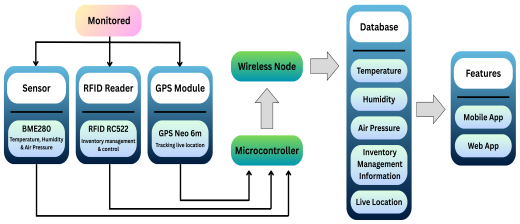

Fig. 1. Block diagram of the proposed system

sensor using the I2C communication protocol to acquire temperature, humidity, and air pressure data. Similarly, the GPS Neo-6M module is connected via UART communication to obtain real-time location coordinates.

A MicroSD card module is attached to the system to store sensor data locally. This acts as a fail-safe mechanism in case of network connectivity issues during transit.

A voltage and current sensor is integrated into the power management system to monitor the battery health and ensure consistent power supply to the modules. A rechargeable Lithium Polymer battery provides portability and ensures uninterrupted operation during transportation.

Thin silicon wires and careful soldering techniques are used for stable and insulated connections, with heat shrink tubing applied at joints to maintain system safety and durability.

The overall schematic design ensures efficient data acquisition, processing, storage, and transmission, supporting the real-time monitoring needs of the supply chain tracking system.

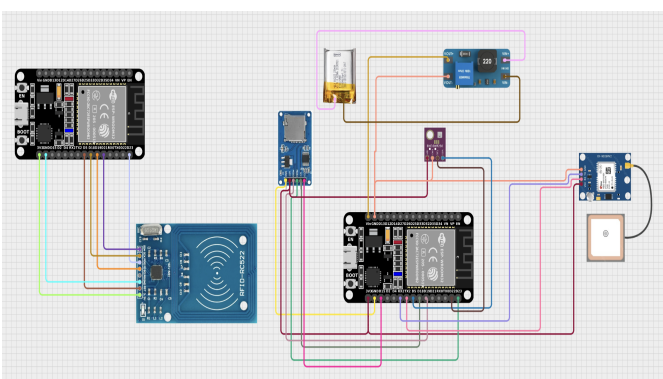

Fig. 2. Circuit diagram of the system

C. System Architecture

The system architecture is divided into five primary layers, as illustrated in Fig. 3:

Farm Layer: At the farm warehouse, agricultural produce is tagged using RFID tags. Environmental parameters inside the storage area are monitored in real-time using the BME280 sensor. The warehouse location is updated using GPS data. The first ESP32 collects sensor data and inventory information, storing it locally and transmitting it to the cloud.

Transport Layer: During transit, the vehicle carrying the goods is equipped with sensors, RFID reader, and a GPS module. Real-time data about the environmental conditions and live vehicle location are transmitted wirelessly to the cloud, ensuring constant monitoring of goods during their journey.

Market Layer: Upon arrival at the marketplace, RFID scanning is conducted again to confirm inventory delivery. The system updates the storage and transport history, ensuring complete traceability.

Cloud Layer: The cloud server acts as a central repository where all collected data is uploaded. It hosts a secure database that stores environmental readings, inventory movements, and live location data. APIs are employed to allow secure access and retrieval of the stored information by the user interfaces.

User Interface Layer: The mobile and web applications provide stakeholders with real-time dashboards displaying environmental conditions, live tracking, and inventory information. Alerts and notifications are triggered if any monitored parameter exceeds predefined thresholds, enabling quick decision-making to preserve the quality of the products.

The layered system architecture ensures seamless interaction between hardware, cloud, and users, promoting an efficient, transparent, and reliable agricultural supply chain management process.

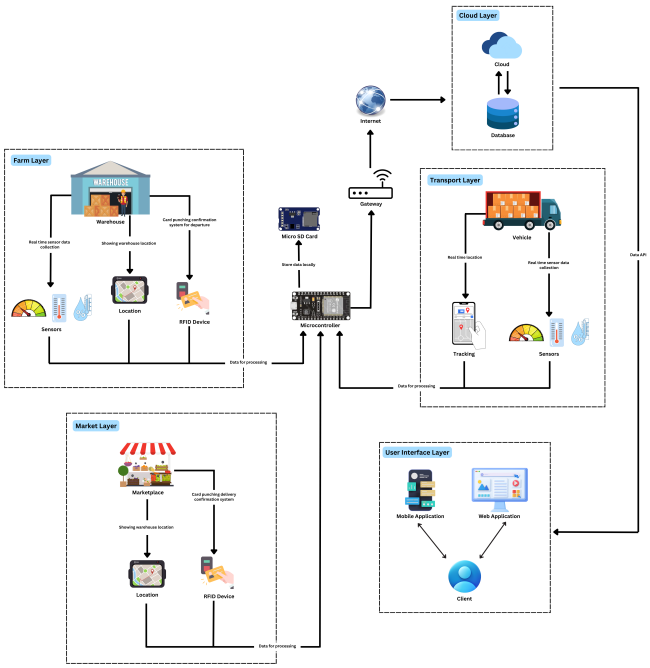

Fig. 3. System architecture

V. Result Analysis

The Agricultural Supply Chain Tracking System developed in this project demonstrates notable effectiveness in enhancing traceability and monitoring environmental conditions during the transportation of agricultural goods. By integrating GPS, RFID, and IoT-based sensors, the system continuously records key data points, allowing for better management of perishable produce across various transit stages.

The values recorded during testing are presented in Table I, summarizing real-time sensor data including geolocation, temperature, humidity, and atmospheric pressure.These readings were taken over an interval of 50 minutes.

TABLE I
SENSOR DATA SUMMARY

Parameter	Value Range (Sample)	Mean	Min	Max
GPS Latitude (Â°)	0.000000, 23.824223, ...	5.81	0	23.82
GPS Longitude (Â°)	0.000000, 90.434494, ...	22.06	0	90.43
Temperature (Â°C)	30.99, 31.19, ..., 28.31	30.41	28.31	31.85
Humidity (	67.31, 67.28, ..., 58.45	62.42	58.45	68.48
Pressure (hPa)	1008.46, 1008.38, ..., 1008.51	1008.43	1008.31	1008.53

Fig. 4.

The dashboard interface was designed for user-friendly, real-time visualization of these parameters, supporting immediate decision-making when deviations occur. This ensures proper handling conditions are maintained throughout the journey—from the farm to the market—thereby reducing spoilage risk and enhancing produce shelf life. As shown in Fig. 4 and 5 (real-time data dashboard), the system efficiently tracks changes in environmental conditions and location in real time. This not only reinforces operational transparency but also empowers farmers, distributors, and retailers with actionable data insights.

Fig. 5.

To evaluate the effectiveness and feasibility of our proposed smart agricultural supply chain tracking system, we conducted a comparative analysis with existing technologies, including RFID-based systems, IoT and WSN systems, blockchain-integrated IoT systems, and chipless RFID sensors. The comparison is summarized in Table II and is based on several key criteria such as cost, sensor support, GPS functionality, scalability, and industrial suitability.

TABLE II
A COMPARISON OF THE SUGGESTED SYSTEM

Feature	Proposed system	RFID Based System	IoT and WSN System	Blockchain and IoT System	Chipless RFID Sensor
Cost (USD)	Low (100-200)	Medium (200-500)	Low (100-300)	High (500-1000)	Low (50-200)
GPS Tracking	Yes(Neo 6M)	No	No	Yes	No
RFID Support	Yes(Card Module)	Yes	No	Yes	No
Temperature Detection	Yes (BME280)	No	Yes	Yes	Yes
Humidity Detection	Yes (BME280)	No	Yes	Yes	No
Pressure Detection	Yes (BME280)	No	Yes	Yes	No
Data Logging	Yes (MicroSD)	No	Yes (Cloud)	Yes (Blockchain)	No
Remote Monitoring	Yes	Yes	Yes	Yes	Yes
Scalability	Moderate	High	Moderate	High	Low
Latency	Low	Low	Moderate	High	Low
Industrial Suitability	Yes	Yes	Yes	Yes	Limited

Our proposed system uniquely integrates GPS (Neo 6M) and RFID (Card Module) technologies with environmental monitoring capabilities using the BME280 sensor, which captures temperature, humidity, and pressure. It also supports local data logging via MicroSD, ensuring resilience during network outages, and offers remote monitoring through cloud integra-

tion.Compared to existing systems: RFID-based systems lack environmental sensors and GPS. IoT and WSN systems offer sensor capabilities but often exclude RFID and are less cost-effective. Blockchain systems provide robust data integrity but are expensive and suffer from high latency. Chipless RFID sensors are low-cost and humidity-specific but have limited industrial adaptability and lack GPS and remote tracking features. Our system demonstrates a balanced trade-off between cost-efficiency, real-time tracking, and scalability, making it especially suitable for deployment in developing agricultural markets where budget and infrastructure are key constraints.

In summary, the proposed Agricultural Supply Chain Tracking System offers a practical, cost-effective, and scalable solution to modernize agricultural logistics. It strengthens quality assurance, improves accountability, and contributes to solving global challenges in food security and agricultural efficiency.

VI. Challenges Faced

The development and implementation of the Agricultural Supply Chain Tracking System encountered several challenges:

GPS Signal Limitations: The Neo-6M GPS module often struggled to establish and maintain satellite connections, especially in regions with dense buildings, heavy foliage, or poor weather. This was exacerbated by the limited number of satellites that regularly pass over Bangladesh. As a result, the system experienced delays in location acquisition and occasional gaps in tracking during transit. These disruptions necessitated the use of fallback mechanisms such as local coordinate logging.

Unreliable Internet Connectivity: Real-time data transmission was frequently hampered by inconsistent or unavailable Wi-Fi connectivity, particularly in rural areas or while the transport was in motion. Since the ESP32 relies on internet access to transmit data to the cloud, this resulted in synchronization delays and occasional data loss. The MicroSD card module was used as a backup storage solution to retain data during offline periods, but it affected the responsiveness of the live monitoring system.

Cloud Synchronization Latency: Even when connectivity was available, transient Wi-Fi dropouts during vehicle movement introduced delays in data updates to the Firebase cloud server. This reduced the real-time visibility of environmental and location parameters and required additional logic to manage data queuing and re-sending.

Despite these challenges, the system remained functional and resilient due to its modular architecture and built-in fail-safes. These limitations underscore the need for alternative communication technologies such as GSM modules or LPWAN for future versions of the system to ensure reliable deployment in low-connectivity regions.

VII. Conclusion

The Agricultural Supply Chain Tracking System developed in this project offers a comprehensive solution for monitoring the movement and condition of perishable agricultural products from the point of origin to the point of sale. By integrating GPS for geolocation, RFID for product identification, and IoT-based environmental sensors, the system enables real-time data acquisition, transmission, and visualization across the supply chain.The implementation demonstrates the potential of smart technologies to address critical challenges such as traceability gaps, spoilage due to poor storage conditions, and a lack of accountability in conventional logistics. The system ensures end-to-end visibility of agricultural goods, allowing stakeholders to monitor temperature, humidity, pressure, and location through a user-friendly cloud interface. Real-time access to this information supports data-driven decision-making and rapid response to deviations from optimal conditions. Despite facing challenges such as GPS inaccuracy in obstructed environments and intermittent network connectivity in rural regions, the modular design and offline data logging ensured uninterrupted functionality and data integrity. The system's resilience and effectiveness underscore its potential for deployment in real-world agricultural settings, especially in developing regions where supply chain inefficiencies are prevalent.

In summary, the proposed solution is cost-effective, scalable, and highly adaptable to various agricultural logistics scenarios. Future enhancements may include GSM or LoRa-based communication modules for improved connectivity and blockchain integration for secure, tamper-proof traceability. The project lays a strong foundation for transforming traditional supply chains into intelligent, transparent, and sustainable systems capable of meeting the growing global demand for food safety and efficiency.

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